

[DC] Perception of Visual Variables on Various Tiled Wall-Sized Display Settings in Immersive Environment

Dongyun Han*
Utah State University

ABSTRACT

Immersive analytics indicates carrying out visual analytics in interactive virtual environments (IVEs). As immersive technologies have rapidly matured with commercial success, researchers have sought to leverage their potential to aid human data understanding and sense-making processes. Unlike the limited information space available on 2D display settings, users in IVEs can use 3D immersive space as information space by creating as many visualizations as they want. Despite growing knowledge of its benefits, many challenges still remain for immersive analytics to be effectively deployed for various applications. Effective use of data visualizations in IVEs requires a better understanding of how people perceive, interpret, and process visualized information. This paper discusses my research questions and presents research progress.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI;

1 INTRODUCTION

Immersive analytics refers to performing visual analytics in interactive virtual environments (IVEs). Immersive technologies such as Virtual Reality (VR) and Augmented Reality (AR) recently has attracted many visualization researchers. Researchers have sought to leverage their potential to aid human data understanding and sense-making processes. Despite growing awareness of the benefits of immersive analytics, many challenges still remain for immersive analytics to be effectively deployed for various applications [3].

Earlier research found that tiled wall-sized displays in the real world have various cognitive and user performance benefits in performing visualization analysis tasks [7]. For example, multi-displays can present an enormous amount of data visualizations at once and could help users with data comparison tasks by providing additional spaces for users' external memory. Some advantages stem from having a broader field of view on the information space [1], which can be easily realized in an IVE without the need to consider the cost of displays or the requirement of physical space for assembly. Immersive analytics allows users to create as many visualizations as they want by using 3D immersive space as their information space for comprehending data. Furthermore, unlike the real-world setting, displays in an IVE can be simply rearranged in different layouts in its immersive space like cylindrical or spherical layouts [5].

To provide users with a better data analytical experience in IVEs, it is essential to gain a deeper understanding of user interactions [6]. The goal of my doctoral thesis is to investigate the effectiveness of user interactions in terms of data navigation in order to enhance human perception in IVEs [4]. In this paper, I discuss my research questions and report research progress to examine how accurately people perceive visual elements across different large display layout settings for immersive analytics. The findings of this research

are expected to enhance our understanding of the effectiveness of immersive analytics.

2 MOTIVATION AND RESEARCH QUESTIONS

To fully utilize tiled wall-sized displays for data analysis, it is critical to understand how human perception is affected by these workspaces. Data analysis tasks include identifying and comparing values encoded in visual elements such as length, area, angle, and slope. In a bar chart, for example, users see and compare the length of the bars and even analyze trends based on their slope. However, in a large display environment, data analytics is accompanied by user position, user orientation, and distance to a displayed visualization, all of which affect its perceived size. The perceived size of an object and its actual size might be different, which is called vision distortion. The same distortion issue would be observed with large display environments in IVEs, and it could be much worse than it is in the real-world due to a variety of factors such as the discomfort of wearing a head mount display (HMD) and the difference in resolution between the human eye and the HMD device.

At the same time, however, IVEs have multiple advantages over the real-world. First of all, the large display layouts in IVEs can be easily adjusted into a variety of arrangements including flat, circular, and spherical shapes. Display heights can also be adjusted by following the heights of users. Furthermore, novel 3D UI interactions in IVEs, such as teleportation and bringing distant displays closer to users, have the potential to improve visual analytics performance and user experience. However, it is unclear which display layout and interaction would result in less amount of distortion to perceive visualization elements. It is also unknown how much distortion exists in IVEs when compared to a real-world display setting. The specific research questions for this proposal are the followings:

- RQ1.** How much distortion do display arrangements in IVEs contribute to the perception of visual elements?
- RQ2.** How much distortion does IVEs have in comparison to a real-world display setting?
- RQ3.** What navigation technique would be the most effective in reducing visual distortion?

3 RESEARCH PLAN

I am planning to conduct two formal user studies to investigate the amounts of vision distortion on perceiving visualization elements that are displayed on large display settings in an IVE. In previous research, Bezerianos and Isenberg [2] and Wigdor et al. [8] investigated vision distortion by user and visualization positions employing large wall and tabletop display environments respectively in the real-world. I design two studies that are as close to the work of Bezerianos and Isenberg [2] as feasible in order to examine the difference in visual distortion between an IVE and the real-world.

3.1 Study 1: Vision Distortion in IVEs

The first study aims to investigate the amount of vision distortion on perceiving visualization elements presented in the main display and wall display layout settings. In this study, participants will be asked to engage in size production tasks while standing in front of a virtual

*e-mail: dongyun.han@usu.edu

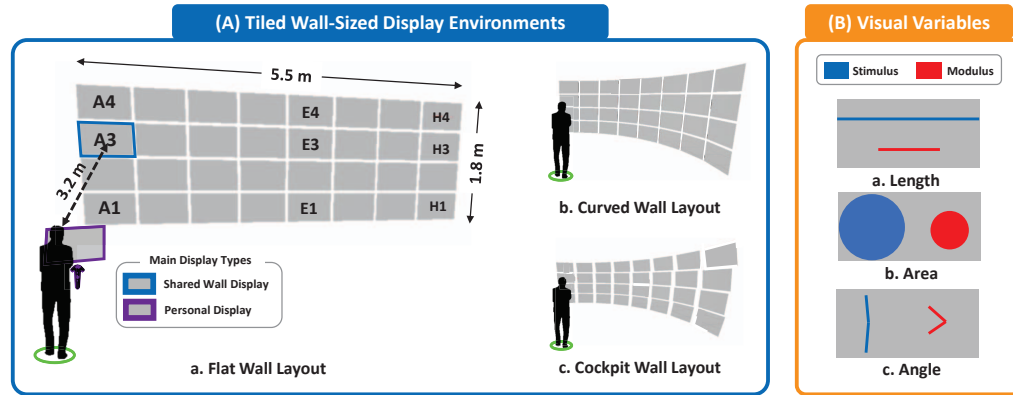


Figure 1: (A) The tiled wall-sized display environments showing their dimensions and modulus locations and (B) the visual variables

large wall display in an IVE, interactively adjusting the magnitude of a visual variable on a main display to match the magnitude of the same variable on another display (i.e., A1 ~ H4 in Fig. 1). The size of a large wall display in this study is about 5.5 m (width) $\times$ 1.8 m (height), with a total of 32 individual displays arranged in 4 rows and 8 columns. Each individual display measures 61 cm (width) $\times$ 41 cm (height) and has 2 cm bezels. Participants will be positioned to the left of the wall display rather than in the center, 3.2 m away from it. This is due to the fact that results are expected to be symmetric left and right. Two types of main display settings are considered. The first main display type is a shared display condition, which is part of the wall display and can be found at A3 in Fig. 1. The second type is a personal display type which is located near a user and follows his or her controller. It is difficult to achieve in the real-world due to the display weight, but it is simple in IVEs. The size of this personal display is identical to the individual displays.

Three display layouts (i.e., *Flat*, *Curved*, and *Cockpit* in Fig. 1-A) are considered in this study. *Flat* aligns tiled displays as a flat wall. *Curved* wraps the displays around a user and rotates their y-axis orientation to face the user. *Cockpit* is similar to *Curved*, but the orientation of the individual displays in this setting is fully rotated in the direction to face the user.

The three visual elements include *Length*, *Area*, and internal *Angle* (Fig. 1-B). The same but different sized two visual elements called a stimulus and modulus, are presented to a participant in each task. The stimulus is shown on the main display, and the modulus is shown on one of the wall displays. The participant has control over the stimulus and is asked to make it as close to the size of the modulus as possible until it is perceived to be equivalent. The task completion time and accuracy will be measured and analyzed to investigate the effectiveness and the amount of vision distortion by following the wall display arranged types, main display types, and modulus location. In order to examine their behavior, participants' head and gaze movements will be recorded during the study.

3.2 Study 2: Vision Distortion and Navigation

The second study is aim to compare interaction techniques to explore data in IVEs. I am planning to conduct the study using *Flat* wall-sized display condition because it is expected to have higher amount of visual distortion than other conditions in Study 1. It will be a within-subject study, with participants performing tasks similar to those in Study 1 but using different techniques. Expected data exploration methods to be investigated are *Standing* and *Moving* (navigating the IVE) for the wall main display type and *Moving* and *Bringing* (bringing a distant display close to the user) for the personal main display type. The baseline condition for this study is *Standing*. Study 2 will allows us to examine which exploration technique

is most effective at reducing vision distortion when performing immersive analytics in a collaborative setting.

4 CONCLUSION

In this paper, I proposed my on-going research work including the research questions and user study plan for my dissertation. My Ph.D. journey began during the pandemic, and I rarely had opportunities to get feedback from VR communities on my research topic. The IEEE VR Doctoral Consortium will be held during my fourth year of Ph.D. studies, and I believe it will a great opportunity to discuss my work and challenges in a broader context.

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REFERENCES

- [1] R. Ball, C. North, and D. A. Bowman. Move to improve: promoting physical navigation to increase user performance with large displays. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 191–200, 2007.
- [2] A. Bezerianos and P. Isenberg. Perception of visual variables on tiled wall-sized displays for information visualization applications. *IEEE Transactions on Visualization and Computer Graphics*, 18(12):2516–2525, 2012.
- [3] B. Ens, B. Bach, M. Cordeil, U. Engelke, M. Serrano, W. Willett, A. Prouzeau, C. Anthes, W. Büschel, C. Dunne, et al. Grand challenges in immersive analytics. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, pp. 1–17, 2021.
- [4] D. Han and I. Cho. Evaluating 3d user interaction techniques on spatial working memory for 3d scatter plot exploration in immersive analytics. In *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2023.
- [5] J. Liu, A. Prouzeau, B. Ens, and T. Dwyer. Design and evaluation of interactive small multiples data visualisation in immersive spaces. In *2020 IEEE conference on virtual reality and 3D user interfaces (VR)*, pp. 588–597. IEEE, 2020.
- [6] K. Marriott, J. Chen, M. Hlawatsch, T. Itoh, M. A. Nacenta, G. Reina, and W. Stuerzlinger. Immersive analytics: Time to reconsider the value of 3d for information visualisation. *Immersive analytics*, pp. 25–55, 2018.
- [7] L. Shupp, R. Ball, B. Yost, J. Booker, and C. North. Evaluation of viewport size and curvature of large, high-resolution displays. In *Graphics Interface*, pp. 123–130, 2006.
- [8] D. Wigdor, C. Shen, C. Forlines, and R. Balakrishnan. Perception of elementary graphical elements in tabletop and multi-surface environments. In *Proceedings of the SIGCHI conference on Human factors in computing systems*, pp. 473–482, 2007.